

The Mortal Scientist

Learning as cost-accounted computation in a reflective higher-order calculus:
hypotheses in namespace logic, assays as guarded receipts,
and an ecology in the gap between a calculus and its image

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Abstract

We ask what it would take for a computation to do science. Learning is modelled here not as fitting a function to a supplied dataset but as the loop the scientific method describes: form a hypothesis, design an experiment that could refute it, run the experiment against the world, read the verdict, and revise. A learner of this kind has no dataset. It must go and get its observations; getting them costs something; and getting them changes the thing observed.

Writing that loop down requires four things. An account of what the learner’s environment is, and of what it could in principle come to know of it. A language in which hypotheses are stated — fine enough to express every distinction the learner could observe, and no finer, since a language that outruns observation invites the learner to spend its life chasing differences no experiment could settle. A mechanism turning a hypothesis into an experiment and an experiment into a verdict. And a motive: some reason the learner prefers better hypotheses to worse ones, which had better not be an external scorekeeper if the model is to stand on its own.

We answer these in order. The environment of a computation is other computations, since computations are ontologically isolated; what a learner can know of one is bounded above by its context-labelled bisimulation class. The hypothesis language is therefore not chosen but generated: we take the logic that the OSLF family of algorithms emits from the presentation of the ρ calculus — namespace logic — whose formulae denote bisimulation equivalence classes, and we admit its structural predicates, which is what equips the learner with senses. Hypotheses are installed operationally as **where**-clause guards on receipts, so that an assay is a receipt and a confirmation is a rendezvous. For the motive we gamify: tokens are conserved and never minted, the learner holds a bounded internal supply, further stacks lie distributed through the environment including inside other computations, and breaking a computation open to harvest its stack is the analogue of eating. Nobody adjudicates; the learner that predicts badly starves.

Three results organise what follows. Because the ρ calculus is Turing complete we may work in the image of the translation of cost-accounted ρ back into ρ , and in that image predation is nothing but a receipt — so the ecology lives precisely in the gap between the cost-accounted calculus and its image, where the erasure stops being faithful against contexts outside the image. Because harvest ends the prey’s computational life, each specimen affords exactly one measurement, and hypotheses about individuals are worthless: the destructive character of the assay *forces* the hypothesis language to be a logic of namespaces. And because a closed computation cannot forage, metabolic viability lower-bounds observability — perfect concealment is available only to the starving. Reproduction is then given by

reflection: a genome is the *quotation* of a scientist's code, inert because quotation does not reduce, and crossover is a fine-grained exchange of parands between two such quotations at homologous contexts. We close with distributions over ecologies, in which science turns out to be a phase rather than a universal advantage; an audit separating what the assumptions force from what they leave free; and an analysis of whether the assignment of ecologies to cost-accounted graph-structured lambda theories is functorial. It is not, quite, and the failure is instructive. A closing section posits — without designing — a perceptual front end encoding the world as a population of such computations, and argues that if the mind is a computational apparatus then an architecture of this shape is close to forced, and that the effectiveness of mathematics is then not unreasonable but structural: the hypothesis language is generated by the same relation that determines what can be observed at all.

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1 Introduction

1.1 The scientist as a learner

Most computational accounts of learning begin with a dataset: a supply of observations that exists independently of the learner, whose acquisition is free, and whose acquisition does not disturb the source. Given such a supply the problem is one of induction — find the hypothesis that best accounts for what is already in hand — and the interesting questions are statistical.

A scientist has none of this. There is no dataset waiting; there is a world, and the scientist must decide what to do to it in order to make it answer. The characteristic activity of the scientific method is therefore not induction but *the design of experiments*: choosing an intervention whose outcome would discriminate between the hypotheses currently in play, performing it, and letting the result decide. Induction is what happens afterwards, and it is the easier half.

Three features of that activity are ordinarily abstracted away and are exactly what we want to keep.

Observation is action. To learn something about a system one must do something to it. There is no passive channel. Every measurement is an intervention, and the system afterwards is not the system before.

Inquiry is expensive. Experiments consume resources that could have been spent otherwise, and a scientist with a finite endowment can run only finitely many. Which questions are worth asking is therefore not a philosophical question but a budgetary one.

The scientist is in the world it studies. It is not a disembodied observer looking on from outside. It is made of the same stuff as its subject matter, subject to the same constraints, and visible to whatever else is looking.

A model that keeps all three will look less like statistics and more like ecology, and that is the model we build.

1.2 What such a model must supply

Before any formalism, here is what has to be on the table.

- (R1) **An environment, and a ceiling on what can be known of it.** One must say what the learner is learning *about*, and — crucially — what the best possible outcome of inquiry would be. Without a ceiling there is no way to say whether a learner is doing well.
- (R2) **A language for hypotheses.** Hypotheses must be stated in something. That something should be able to express every distinction the learner could in principle observe, and no more. A language too coarse leaves the learner unable to say what it can see; a language too fine sends it chasing differences that no experiment could ever settle.
- (R3) **A mechanism from hypothesis to verdict.** There must be an account of what an experiment *is*, how a hypothesis determines one, and how the outcome comes back as confirmation or refutation — including an honest account of the case where nothing comes back at all.
- (R4) **A motive.** Something must make the learner prefer better hypotheses. If this is an external scorekeeper handing out rewards, the model has outsourced the very thing it set out to explain, and one may reasonably ask who scores the scorekeeper.

1.3 Four answers

Our answers are as follows, and the remainder of the introduction says why each is less a choice than it appears.

(R1) The environment is other computations. We take as premise that computations are ontologically isolated: a computation has no access to anything but through interaction. It follows immediately that the environment of a computation is other computations, and that what one can come to know of another is bounded above by the equivalence induced by the contexts one can build against it — context-labelled bisimulation [4]. That is the ceiling R1 demands, and it is derived rather than posited. We fix the model of computation as the ρ calculus [10].

(R2) The hypothesis language is generated, not designed. Given the ceiling, R2 has a canonical answer: the language must be adequate for context-labelled bisimulation, which is exactly what a logic generated from the calculus’ own presentation gives. Feeding the ρ calculus’ presentation to the OSLF family of algorithms [11] emits namespace logic [9], whose formulae denote bisimulation equivalence classes. We admit the whole of it, including the structural predicates — and admitting those is what gives the learner senses (§3).

(R3) An assay is a guarded receipt. We work in the variant of the ρ calculus whose for-comprehension carries a **where** clause [3, §9], so a formula may sit in guard position on a receipt. A hypothesis installed as a guard *is* an experiment, and a confirmation is a rendezvous. The checker is the reduction rule; nothing is interpreted, and the test is metered by the same apparatus that meters everything else.

(R4) The motive is metabolic. Cost accounting as a monad [5] makes the learner mortal: every step is paid for, and a computation that cannot pay stops. We then gamify. Tokens are conserved and never minted; the learner holds a bounded internal supply; further stacks lie distributed through the environment, including inside other computations; and breaking a computation open to harvest its stack is the analogue of eating. This is the answer to the scorekeeper problem. A hypothesis says *where the tokens are*, and the reward for a good one is the tokens themselves. Predictive accuracy and metabolic success become the same quantity, measured in the same units, with nobody adjudicating.

That is the whole of what we put in. The rest of the note is largely the discovery that we had no choices left: §14 audits which features are forced by this list and which remain free, and the list of free parameters is short.

1.4 What we are claiming, and what we are not

We are claiming that the commitments just listed determine almost all of the apparatus one would otherwise have to invent. The interesting sections below are the ones in which we find we had no choices left.

We do treat learning, and §6 is where. What we give there is the geometry of the hypothesis space and the constraints any revision policy must satisfy in it — that the space is an ultrametric tree and so admits no gradient, that incremental revision provably cannot leave the ball it begins in, that the cost of a move is monotone in its distance, and that the branching a searcher can survey is bounded by its wealth — together with a catalogue of strategies compatible with those constraints. We are *not* claiming to have fixed a policy, proved a convergence theorem, or said anything about statistical efficiency. Which policy is best is an ecological question and depends on the distribution of §13; the policy itself remains one of only three genuinely free parameters we find. We do treat reproduction (§12), but we do not treat inheritance of acquired characteristics: whether a lineage’s transmitted material can carry what an individual learned is left open, since the empirical case against a strict germ–soma barrier [21] is strong enough that we would rather not build on either answer.

We are also not claiming a novel logic. The logic is the one obtained for free from the encoding, in exactly the sense of [2]: given the presentation (Σ, E, R) of the rho calculus, the OSLF algorithms emit a logic, and it is the same logic whether one is classifying knot diagrams or hunting for tokens. What is new here is the reading, the game, and three consequences of the game that we did not anticipate.

1.5 Why adequacy is the right demand

R2 asked for a language neither too coarse nor too fine, and it is worth saying why that pair of failures is the right pair to worry about, since adequacy is a technical condition that can look like a formality.

A learner separated from its environment by an interaction cut has exactly one channel of access: what interaction reveals. If its hypothesis language is too coarse, there are distinctions it can observe but cannot state, and it is blind in a way its own faculties do not require. If the language is too fine, it will form hypotheses that differ on environments no experiment could separate, and will spend tokens — which by R4 are its life — discriminating between what are, for it, the same world. Adequacy rules out both at once: the formulae separate precisely what interaction separates.

This is why the logic must be generated from the same presentation that generates the calculus rather than designed. A designed logic has no reason to be adequate. A generated one is adequate whenever its target specification carries the connectives adequacy needs [2, §1.3], and when it does not, the failure is legible and located.

1.6 Ideal logic, checkable fragment, mortal scientist

We inherit the methodological commitment of [2]: the *ideal* logic, adequate for the idealised calculus, is what defines the invariants; *fragments*, restricted so that model checking terminates, are what decide them. In the present setting this distinction stops being methodology and becomes the subject matter. A mortal scientist cannot afford the ideal logic. What fragment it can afford is fixed by its budget, and the fragment it can afford *is* its inductive bias. We return to this in §5, and it is the reason the note’s central objects are approximants rather than limits.

2 Isolation, the cut, and what a computation can know

2.1 The premise

Computations are ontologically isolated. Therefore the environment of a computation is other computations.

We take this as a premise rather than argue for it. Its immediate formal content is that a world factors across an interaction cut. In an interactive graph-structured lambda theory [3], an interaction cut K separates a program from its environment: application for the lambda calculus, parallel composition for CCS, the pi calculus, and the rho calculus. We write

$$W \equiv S \mid E$$

for a world factored into a scientist S and its environment E . Nothing here privileges S : the factorisation is a choice of which side one is standing on, and in §10 we will take seriously that E contains further scientists for whom S is environment.

2.2 Three grades of access, and only three

The premise has a sharp consequence that the rest of the note leans on. S does not hold E ’s term. It cannot read E ’s structure by inspection, because inspection is not an operation the calculus provides across a cut. What it has instead is exactly three things, and it is worth being explicit that the calculus offers no fourth.

Perception. S may evaluate structural predicates at the *interaction surface* — the outermost interaction structure of E , the part exposed at the cut. This is what senses are: one sees the skin of a rock, not its interior.

Interaction. S may build a context K , plug E into it, and observe. This is what the context-labelled modality $\langle K \rangle \varphi$ says, and it is the only route to behaviour.

Reflection. If S *holds* a name, it may look inside it, because in a reflective calculus a name is a quoted process and the name predicate $@\varphi$ sees into that quotation without communicating at all.

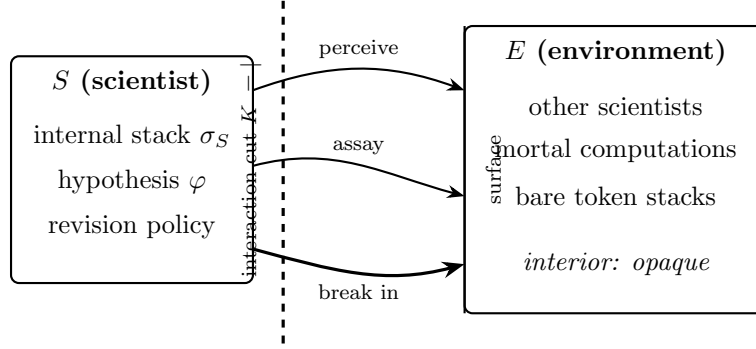


Figure 1: The cut. Perception evaluates structural predicates at the surface; an assay reaches across and perturbs; a break-in passes the surface and is fatal to what lies behind it. There is no fourth route, and the three are ordered by both what they reveal and what they destroy (§9).

The third is the interesting one and it is available only for names S has come to hold. This gives the distinction the note turns on: **observation versus disclosure**. Behaviour is what one observes across a cut; structure is what one is given, or takes. Perception reads the surface of things one does not hold; reflection reads the interior of things one does. The operation converting the second into the first is the subject of §7, and it is fatal to the thing converted.

3 The logic, and senses

3.1 What the generator supplies

We recall only what we use, following [2, §2]. The ρ calculus has term formers 0 , $*n$, $n!(Q)$, $\text{for}(x \leftarrow n)P$, parallel composition, and $@P$; parallel composition carries an associative–commutative equational theory and there is a single communication rewrite. Feeding this presentation to the generator yields:

Structural connectives. One connective per term former. We write $\text{out}(n)\varphi$ for the predicate holding of $n!(Q)$ with $Q \models \varphi$, and $\text{in}(n)\varphi$ for the analogue on receipts. Because $|$ is associative–commutative, the generated connective for parallel composition is a separating conjunction:

$$P \models \varphi \mid \psi \iff P \equiv Q \mid R \text{ with } Q \models \varphi, R \models \psi.$$

Behavioural modalities. One primitive modality $\langle K_j \rangle$ per rewrite rule and choice of redex position, labelled by the one-hole context in which the step fires.

Name predicates. A name is a quoted process, so

$$n \models @ \varphi \iff n = @P \text{ with } P \models \varphi,$$

which is namespace logic [9]: one describes a namespace by a property rather than enumerating it. §8 shows this is not a convenience but a requirement.

Restriction, fixed point, grading. The hidden-name quantifier $P \models \mathbf{H}x.\varphi$ iff $P \equiv \mathbf{new} \ x \ \mathbf{in} \ Q$ with $Q \models \varphi$ and x fresh; the greatest fixed point $\nu X.\varphi$ with X positive; and graded separation $\varphi \mid_k \psi$, which splits a term under $\mathbf{H}$ with exactly k shared restricted names [2, §2.3].

Remark 1 (Restriction is sugar). We treat **new** as syntactic sugar throughout. In a reflective calculus a name is a quoted process, so a “fresh” name is manufactured by a structural scheme rather than drawn from a stock of atoms, and freshness is relative to a scope rather than global. We remain agnostic about the scheme, subject to one constraint: it must be *decentralised*, manufacturing names from material a computation already holds rather than by consulting a registry. That constraint is not an engineering preference. A central name server would be an oracle sitting across the cut and consultable by every computation, which contradicts the isolation premise of §2 outright.

Two consequences run through the rest of the note. Names have *structure*, so a name predicate sees all the way into one and the hidden-name quantifier is descriptive rather than opaque; and names have *size*, so by Definition 3 they carry a running cost. Concealment is therefore never secrecy, only price — which is why δ was introduced as a cost parameter and not as a wall.

3.2 The structural predicates are admitted, and they are senses

In [2] the structural predicates are wielded by an analyst standing outside the term, holding it in hand, free to count its gates and test its decompositions. That is not the scientist’s situation, and admitting the structural predicates without saying where they may be applied would hand the scientist an oracle and destroy the isolation premise of §2.

The discipline that saves them is that a sense is a structural predicate evaluated *at the surface*.

Definition 2 (Depth). The *depth* $d(\varphi)$ of a structural formula is the maximal nesting of term-former connectives and prefixes occurring in it, with $d(\top) = 0$, $d(\text{out}(n)\varphi) = d(\text{in}(n)\varphi) = 1 + d(\varphi)$, $d(\varphi \mid \psi) = \max(d(\varphi), d(\psi))$, $d(\mathbf{H}x.\varphi) = 1 + d(\varphi)$, and $d(@\varphi) = d(\varphi)$.

Definition 3 (Sensing, and its price). For $W \equiv S \mid E$, the scientist S may evaluate φ against E whenever $d(\varphi) \leq \alpha$, where α is S ’s *acuity*; the evaluation debits S ’s stack by $\kappa_{\text{sense}}(d(\varphi))$, a monotone schedule. Acuity is not a fixed endowment: α is whatever S can currently pay for.

Two comments. First, the depth grading is what keeps perception from being an oracle: seeing deeper costs more, so the interior of E is not free, it is merely expensive, and past a point unaffordable. Second, $d(@\varphi) = d(\varphi)$ rather than $1 + d(\varphi)$ is deliberate and records the third grade of access: opening a name one holds is not a passage through a surface, because one already holds it. Reflection is cheap for exactly the reason predation is not.

4 Hypotheses, assays, and the where clause

4.1 The embedding, in operational position

An embedding of the logic into the calculus is what lets **rho** computations compute over hypotheses. The obvious construction — reify formulae as data, write an interpreter — works but is unsatisfying, because the scientist must then carry the interpreter and the interpreter is not

grade	instrument	applies to	price
perception	out, in, $ $, $ _k$, $\mathbf{H}$	the surface of E	$\kappa_{\text{sense}}(d)$, monotone in depth
interaction	$\langle K \rangle \varphi$	the behaviour of E	one rendezvous per step, and time
reflection	$@\varphi$	names S holds	a computation on a name; no communication

Table 1: The three grades of access. The ladder is also an ordering by invasiveness (§9), and in the opposite direction.

part of the theory. The variant of `rho` whose for-comprehension carries a **where** clause [3, §9] does better. Its general form is

$$\text{for}(p_{11} \leftarrow c_{11} \ \& \ \cdots ; \cdots ; p_{1n} \leftarrow c_{1n} \ \& \ \cdots \ \text{where } \vartheta) P$$

with `&` joining within a group, `;` separating groups, and ϑ a condition drawn from the booleans over ground formulae, spatial connectives, and the context-labelled modality, in which the variables bound by the preceding clauses may be mentioned. A guard is a formula in operational position. Hence:

A hypothesis installed as a guard is an experiment, and a confirmation is a rendezvous.

The checker is the reduction rule. Nothing is carried, nothing is interpreted, and the cost of testing a hypothesis is metered by the same apparatus that meters everything else — which is essential here, since an unpriced hypothesis test would be an oracle and would collapse the entire motivational story of §7.

4.2 The assay

Definition 4 (Assay). An *assay* is a pair (K, φ) of a probe context K and a predicted property φ , realised on a probe channel p as

```
K[ ... ] // the probe: perturb the environment
| for (x <- p where phi(x)) { confirm!(*x) } // prediction holds
| for (x <- p where ~phi(x)) { refute!(*x) } // prediction fails
```

The negation in the second guard is what makes this more than an expression of hope. In a concurrent setting a guard that simply fails to fire is indistinguishable from a guard still waiting, so a naive encoding of falsification cannot tell refutation from patience. The complementary pair repairs this.

Proposition 5 (Trichotomy). *Let $A = (K, \varphi)$ be an assay on probe channel p , with φ decidable on the received term within the ambient budget. Then for each message arriving on p exactly one of the two guards is enabled, and the assay yields one of three outcomes: CONFIRM, REFUTE, or $\perp$, where $\perp$ occurs exactly when no message arrives on p before the budget for A is exhausted.*

Proof sketch. φ and $\neg\varphi$ partition the terms on which both are decidable, so at most one guard is enabled per message and the two receipts do not race. If a message arrives and the budget suffices to decide φ on it, one guard fires. Otherwise no output occurs on either **confirm** or **refute**, which is the residual case. $\square$

We stress that $\perp$ is not a defect. It is the honest residue: the environment did not answer, and no amount of logical apparatus can distinguish that from an environment that would have answered later. What it does is separate that genuine ignorance from falsity, which the naive encoding conflates.

4.3 Refutation is budget-relative, and that fixes the fragment

Proposition 6 (Asymmetry of the modalities). *Let $\varphi = \langle K \rangle \psi$. Then CONFIRM is a finite witness for φ , while no finite run witnesses $\neg\varphi$. Dually for $\varphi = [K]\psi$: refutation is finite, confirmation is not.*

The consequence is the note’s first structural fact about mortality. A scientist with a finite budget can only run assays whose outcome is decided within that budget, so the fragment of the ideal logic it can actually test is the one bounded in modal depth and fixed-point unfolding. That fragment is not adequate — a logic whose model checking terminates never is, for a Turing-complete host [2, §1.3]. Therefore:

A mortal scientist necessarily holds a non-adequate theory, and which non-adequate fragment it holds is fixed by what it can pay for. The budget is the inductive bias.

This is not a limitation we regret. It is the formal content of the observation that finite creatures have finite theories, and it locates the inductive bias somewhere unusually concrete: not in a prior, not in a regulariser, but in a token stack.

5 The hypothesis space is the relaxation lattice

We do not need to invent a space of hypotheses. [2, §3.3] constructs one and calls it something else.

Recall the construction. For a process P , a *characteristic formula* χ_P is satisfied by exactly the processes equivalent to P ; it turns an equivalence question into a model-checking question. Beneath χ_P sits a lattice of progressively weaker formulae obtained by forgetting parts of it — the recursion, the continuations, a private name — each forgetting yielding a coarser predicate that more processes satisfy. In [2] this lattice is where a classifier’s dials live. Here it is the space the scientist moves in.

Definition 7 (Revision). A *revision* is a move on the relaxation lattice: *generalisation* forgets a conjunct and moves down, *specialisation* restores one and moves up. A scientist’s trajectory is a walk $\varphi_0, \varphi_1, \varphi_2, \dots$ on this lattice.

Definition 8 (Stratified distance). For φ, ψ agreeing on all approximants below stage n and differing at n , set $d(\varphi, \psi) = 2^{-n}$, where the stages are the ν -unfolding approximants $\nu^0 \supseteq \nu^1 \supseteq \dots$. This is an ultrametric.

The stratification is not chosen for convenience: it is how bisimulation is already approximated, it grades with modal depth, and by Proposition 6 modal depth is what the budget bounds. Distance, depth, and cost are therefore one graded structure rather than three, which is the main reason to prefer this metric to a measure-theoretic one on denotations.

Proposition 9 (The limit of inquiry is unaffordable). *Let E be an environment whose behaviour is not eventually structural in the sense of [2, Rem. 6] — that is, whose recursion follows the reduction rather than a finite structural cycle. Then χ_E is not exact at any finite unfolding, and no scientist with a finite token supply can install it as a guard.*

So χ_E — the complete theory of the environment — sits at the top of the lattice as a limit the scientist approaches and never reaches. Science converges *along* the lattice.

Remark 10 (When a finite theory is complete). The converse case is worth naming because it is the good one. [2, Rem. 6] observes that a recursion following the *structure* rather than the reduction is exact at a finite unfolding: for a diagram with $2n$ arcs, $\nu^{2n} = \nu$ on the nose. Read for the scientist, this says that a finite theory is complete exactly when the environment’s regularity is structural. Recursions that follow the reduction are the unbounded ones. This is a serviceable formal criterion for which subject matters admit closed sciences and which do not, and we suspect it is more useful than it looks.

6 Searching the lattice

Section 5 supplies a space of hypotheses and a metric on it, and says that a scientist’s trajectory is a walk. It does not say how the walk is chosen. That is the revision policy, which we left open as one of the note’s three free parameters (§14), and we leave it open still. What we do here is narrower and, we think, more useful: the metric is an *ultrametric*, and ultrametric spaces are geometrically unlike the spaces search algorithms are usually designed for. Much of what one would reflexively try is unavailable, and one thing one might not have thought necessary turns out to be forced.

6.1 The geometry is a tree

Three facts about ultrametric spaces do the work. Balls are nested or disjoint, never partially overlapping. Every point of a ball is a centre of it, so no point in a neighbourhood is distinguished. And there is no betweenness: for $\varphi \neq \psi$ there is no formula standing partway from one to the other in the sense a convex combination would.

Remark 11 (There is no gradient). Gradient methods are not merely a poor fit here; they are not well typed. A gradient presupposes a direction in which one may move by an arbitrarily small amount and a notion of interpolation between neighbouring points, and an ultrametric space supplies neither. What it supplies instead is branching, so the search available on a hypothesis lattice is tree search, and the strategies below are all tree strategies.

This has a consequence which we regard as the central fact governing revision policies.

Proposition 12 (Confinement). *Let $\varphi_0, \varphi_1, \dots, \varphi_k$ be a revision walk with $d(\varphi_{i-1}, \varphi_i) < 2^{-n}$ for every i . Then $d(\varphi_0, \varphi_k) < 2^{-n}$: the walk never leaves the ball of radius 2^{-n} in which it began.*

Proof. By the ultrametric inequality $d(\varphi_0, \varphi_k) \leq \max_i d(\varphi_{i-1}, \varphi_i)$, and the maximum is below 2^{-n} by hypothesis. $\square$

In a Euclidean space, sufficiently many small steps reach anywhere; here they reach nothing outside the ball they started in, however many are taken and however long the scientist lives.

A policy of incremental revision is not slow at escaping a bad basin — it cannot escape one at all. Whatever ball a scientist’s early commitments place it in is the ball it will die in, unless it makes a move that is not small.

We take this up again in §12, because it turns a biological mechanism into a computational necessity.

6.2 The two parameters

Definition 13 (Revision move). A *revision move* is a pair: a *locus* in the formula tree — which structural predicate is to be varied — together with an *operation* at that locus, weakening or strengthening. The locus fixes the direction; its depth fixes the distance.

So the two parameters the policy must choose are exactly the two the metric grades.

Distance is backtracking depth. A move of distance 2^{-n} agrees with the current hypothesis on every approximant below stage n and differs at n . Small distance means a deep backtrack and a fine adjustment; large distance means a shallow backtrack and the abandonment of an early commitment.

Direction is a choice of sibling. At a node of the tree the available directions are the one-step refinements the generated logic admits at that locus. The branching factor is therefore fixed by which connectives OSLF emitted, and is not ours to choose.

The price schedule follows without further assumption.

Proposition 14 (Cost is monotone in distance). *Evidence gathered by an assay of modal depth k bears on the approximants at stage k and above. A move of distance 2^{-n} therefore invalidates exactly those confirmations obtained at depth n or greater, and the count of invalidated confirmations is non-decreasing as the distance of the move increases.*

A radical revision is expensive in precisely the currency the scientist has been spending: it discards the assays that bought the commitments it abandons. Cheap moves are confined (Proposition 12) and unconfined moves are dear, which is the trade-off any policy is negotiating.

Remark 15 (Two gradings, one poset). The relaxation lattice and the ultrametric grade the same partial order differently, and conflating them is easy. The lattice orders by logical strength: how much of a characteristic formula has been forgotten. The metric orders by depth: how far down the forgotten material sat. Forgetting a conjunct at stage 1 and forgetting one at stage 5 are both a single lattice step, and are metrically very far apart. Definition 13 keeps the two apart by naming the locus and the operation separately.

6.3 Strategies

We catalogue rather than recommend, since which policy is best is an ecological question and depends on the distribution of §13.

Conservative. Always move the least distance that the last refutation forces. Cheap by Proposition 14, low variance, and by Proposition 12 permanently confined to its initial ball.

Radical. On refutation, abandon a commitment high in the tree. Unconfined, and expensive in exactly the confirmations it throws away.

Adaptive, with refutation rate as temperature. Jump far when refutations are frequent — a high refutation rate is evidence that a commitment high in the tree is wrong — and refine when they are rare. This is annealing-shaped, but the temperature is not a schedule imposed from outside: it is the ratio of **refute** to **confirm** events, which the assays of §4 already produce as messages. We regard this as the default policy a designer would reach for, precisely because its control parameter is already an observable of the system.

Direction selection is the second half of the policy, and the candidates differ in what they optimise.

Blind. Choose a sibling uniformly. Requires no evaluation, and is what a lineage does when the choice function of Definition 52 is unbiased.

By formula cost. Prefer siblings whose formulae are cheap to check. This is the minimum description length bias of §16 appearing as a search heuristic rather than as an accounting identity, and it has the merit that the scientist prefers hypotheses it can afford to test.

By discriminating power. Prefer a sibling ψ for which some assay separates ψ from the current φ — that is, design the crucial experiment first and let it select the direction. This is the distinguishing-formula problem for bisimulation inequivalence [12], and it is the most classically scientific of the options: it maximises the information a single assay returns.

By expected yield. Prefer siblings which, if true, locate more tokens.

Remark 16 (Yield, not truth). The last of these is the one this world actually selects for, and it is worth saying so plainly. A mortal scientist is not rewarded for holding true hypotheses but for holding nutritive ones, so a lineage under selection will drift toward yield-biased direction choice and away from discrimination-biased choice wherever the two disagree. This is the pragmatist objection of Remark 80 arriving at a specific mechanism rather than as a general caveat: it is not that the scientist is indifferent to truth, but that its search is steered at every node by what pays.

6.4 The search is itself metered

One further constraint is easy to overlook and is not a detail. Choosing a direction requires evaluating candidate siblings, and evaluation is model checking, and model checking is priced. A scientist therefore cannot survey the branching it faces.

Proposition 17 (Bounded branching). *The number of siblings a scientist can evaluate before committing to a move is bounded by its stack divided by the cost of checking one. Hence the effective branching factor of its search is a function of its wealth, not of the logic.*

Corollary 18 (Poverty is confining). *A scientist whose stack affords few evaluations per move is likelier to take the cheapest available move, and by Proposition 14 the cheapest moves are the shortest. By Proposition 12 a policy of short moves cannot leave its ball. Impoverishment therefore tends to confinement, and the confinement is not recoverable by patience — only by a subsequent move that the scientist could not afford to evaluate.*

We note without developing it that this closes a loop with §11: a source-feeder whose inquiry converges accumulates the surplus that would buy a wide survey, and has nothing left to search for; a prey-feeder that needs the survey is the one paying an unbounded epistemic rent.

6.5 What escapes

Proposition 12 leaves an individual scientist with exactly one route out of a bad basin: a move that is not small, at a cost given by Proposition 14, chosen from a branching it cannot afford to survey. That is a poor prospect, and it is the reason §12 is not an appendix to this note but a part of its argument. A crossover at a locus high in the term tree is, transported along the characteristic-formula construction, a shallow re-branch in the formula tree — a displacement of exactly the kind an incremental walk provably cannot make, and one no individual pays for, because the cost is borne by the offspring that does not yet exist. The escape is not unconditional: by Remark 57 how far a swap can displace is fixed by the recombination discipline, and the disciplines that displace furthest are the ones that kill most offspring.

7 The game

7.1 Conservation

Tokens are conserved and never minted. Write Θ for the total supply of the world. The cost monad’s ledger law and the history monad’s record together give, globally,

$$\sigma + \kappa = \sigma_0 = \Theta,$$

where σ is the total still held in live stacks and κ the total spent. Spent tokens do not vanish; they migrate into the record. So the free supply is monotonically non-increasing, the accumulation of κ is the arrow of time, and there is no primary producer. The whole game is a finite race.

Remark 19 (Science is entropy production). The aggregate effect of every scientist in the world is to convert free tokens into history. This is the precise form of the observation that the cost of remembering is the cost of doing science: the record is made of spent tokens. It also means predation is not profit in any global sense — it is the purchase of someone else’s remaining time.

7.2 Working in the image

Because ρ is Turing complete, cost-accounted ρ translates back into ρ : there is an internalising encoding $\llbracket - \rrbracket : \mathcal{C}(\rho) \rightarrow \rho$ that represents the located token stacks in the undecorated calculus [3, §6]. Without loss of generality we may therefore avail ourselves of the cost-accounted syntax where that is convenient and of the channels at which token stacks are located where that is convenient.

Definition 20 (Metabolic channel). Under $\llbracket - \rrbracket$, a located stack becomes a message at a name. We call that name the *metabolic channel* m_P of the computation P , and we call

$$m_P!(\sigma) \mid \text{for}(t \leftarrow m_P)\{\text{debit}; m_P!(\sigma') \mid \dots\}$$

the *reclaim–debit–reissue* cycle. A mortal computation is a term equipped with a metabolic channel executing this cycle, with a strictly positive burn rate.

Remark 21 (Metabolism is a race condition). Living requires exposing one’s stack as a message, if only for the instant between reclaiming and reissuing. Computing is therefore racing others to one’s own stack. We did not put this in; it is what the comm rule does with Definition 20.

Definition 22 (Harvest, death). A *harvest* of P by S is the receipt $\mathbf{for}(t \leftarrow m_P)$ performed by S . It is an ordinary communication and requires no new rewrite rule. P *dies* when it cannot fund its next step: the linear receipt consumed the message it was going to reclaim.

So eating is a receive, death is starvation, and defence is restriction. All three are already in the calculus.

Remark 23 (One ledger). The translation is not cost-free: simulating the metering costs ρ -steps. We therefore adopt one metering convention throughout and state it once. *Costs are always read in the decorated calculus $\mathcal{C}(\rho)$; the image is used only for the mechanics of transfer.* This matters because the central inequalities of §9 and §13 are statements about ratios of costs, and a non-uniform translation overhead would move their boundaries. It is the same discipline of keeping levels distinct that governs running $\mathcal{C}$ on an $\mathcal{H}$ -equipped model.

7.3 Predation is the failure of internalisation to be faithful

Here is the observation that we take to be the note's chief structural contribution.

In $\mathcal{C}(\rho)$ a located stack is inviolable. That is not an accident of presentation; it is precisely what the located-stack discipline and the ambient-authority fix were built to guarantee. No rewrite takes your stack, because the stack is not the sort of thing a rewrite addresses.

In the image $\llbracket \mathcal{C}(\rho) \rrbracket \subseteq \rho$ the stack is a message at a name, and any process holding that name may receive it.

These are not the same situation, and the encoding is not faithful against arbitrary ρ contexts. It is faithful against *images of $\mathcal{C}(\rho)$ contexts* — which is exactly the refinement of [4]: an encoding may only be probed by the encodings of source contexts, since arbitrary target contexts see strictly more and would never permit bisimulation preservation. Therefore:

Theorem 24 (Predators are non-image contexts). *Let $\llbracket - \rrbracket : \mathcal{C}(\rho) \rightarrow \rho$ be the internalising encoding and let $C[-]$ be a ρ context. If C lies in the image of $\llbracket - \rrbracket$ on contexts, then C cannot harvest: the stack of any $\llbracket P \rrbracket$ plugged into C is preserved, since the source discipline forbids the corresponding step. If C lies outside the image and holds m_P , it may harvest. Hence the predatory contexts are exactly the non-image contexts holding a metabolic name.*

Proof sketch. Faithfulness of $\llbracket - \rrbracket$ against image contexts is the hosting condition of [4]: every step available to $C[\llbracket P \rrbracket]$ with $C = \llbracket C' \rrbracket$ is the image of a step of $C'[P]$ in $\mathcal{C}(\rho)$, and no step of $\mathcal{C}(\rho)$ removes a located stack from a term. A context outside the image is under no such constraint; a bare receipt on m_P is such a context and takes the stack. $\square$

The decorated calculus is the world in which property rights hold. Its image is the state of nature. The ecology lives in the gap, and eating is what the erasure theorem's side condition was protecting against.

This gives the game a rulebook rather than a stipulation.

Definition 25 (The game). A *game* is a choice of admitted context class $\mathcal{A}$ with $\llbracket \mathcal{C}(\rho)\text{-Ctx} \rrbracket \subseteq \mathcal{A} \subseteq \rho\text{-Ctx}$.

The two endpoints are degenerate: $\mathcal{A} = \llbracket \mathcal{C}(\rho)\text{-Ctx} \rrbracket$ is decorated cost-accounted ρ with no game at all, and $\mathcal{A} = \rho\text{-Ctx}$ is total war in which any context may take anything it can address. The interesting point is the middle one:

Definition 26 (Capability gating). The *capability-gated* game admits those contexts that *derive* m_P from structure they can perceive, and excludes those that are simply handed it.

Since names are unforgeable, derivation is real work, and it is the work namespace logic does: describing a namespace by a property rather than enumerating it. Under capability gating, foraging is name discovery and a hypothesis is a namespace description that either does or does not locate real tokens. Everything below is stated for the capability-gated game.

7.4 The foraging inequality

Conservation has an immediate consequence which we will lean on repeatedly: harvest is not automatically profitable, because finding and opening a stack costs tokens drawn from the same conserved supply.

Proposition 27 (Foraging inequality). *A harvest of P by S credits S on balance only if*

$$\sigma_P > \kappa_{\text{sense}}(\delta_P) + \kappa_{\text{assay}} + \kappa_{\text{break}},$$

where σ_P is the stack recovered and δ_P the depth at which m_P lies concealed.

Corollary 28 (The profitable band). *Since a well-stocked computation can afford deeper concealment — raising κ_{sense} — while a poorly stocked one offers little σ_P , viable prey occupy a band: too poor is not worth opening, too rich is not worth cracking.*

This is optimal foraging theory, and we did not import it; it is what Proposition 27 says once concealment is priced by Definition 3. It is also the inequality whose sign change draws the phase boundaries of §13.

8 Destructive assay forces a logic of namespaces

Harvest ends the prey’s computational life. That single decision has a consequence for the hypothesis language which we did not expect and which we now regard as the best available answer to “why namespace logic.”

Proposition 29 (One specimen, one measurement). *Let P be a mortal computation and let S harvest it. Then no further assay may be run against P . Consequently a hypothesis whose subject is the individual P has zero expected yield after its own first test.*

Proof sketch. By Definition 22 the harvest consumes m_P ’s message and P cannot fund a further step, so P contributes no further transitions and no further surface. A hypothesis φ whose denotation is $\{P\}$ is therefore, after the test, a predicate with empty extension among live computations; whatever it cost to form and hold, it cannot be recouped. $\square$

Corollary 30 (Induction is forced). *In an ecology with destructive assay, the only hypotheses with positive expected yield are those whose subject is a class of computations. The hypothesis language must therefore be able to describe a collection by a property rather than by enumeration.*

That is namespace logic’s definition. So the logic is not adopted for elegance or for uniformity with a companion note; it is what survives the game. Induction — the move from the specimen to the kind — is not an epistemic virtue the scientist is asked to display but the only strategy that pays when each specimen affords exactly one measurement.

Remark 31 (The biological reading). Destructive assay is not exotic. Large parts of experimental biology sacrifice the animal, and the inferential structure of those fields is exactly the one Corollary 30 describes: because you cannot re-measure the individual, everything must be said about the strain, the cohort, or the genotype. The formal setting reproduces the methodological consequence, which we take as evidence that the modelling is not merely decorative.

9 Exposure, and the price of being known

9.1 The exposure lemma

The prey’s optimal defence in a world of predators looks, at first, unbeatable. Restrict everything. A fully closed computation is a live τ -network with no free names; no context can communicate with it; and by the pessimism of [2, §1] its observable behaviour carries almost no information. It cannot be perceived, so its metabolic channel cannot be derived, so it cannot be harvested.

It also cannot eat.

Lemma 32 (Exposure). *Let P be a mortal computation with strictly positive burn rate and no free names. Then P performs no harvest, its stack is monotonically decreasing, and P dies in finitely many steps.*

Proof sketch. Harvest is a receipt on a metabolic name m_Q belonging to some Q outside P (Definition 22). By Remark 1 there is no primitive restriction to appeal to, so the hypothesis must be read structurally: P is closed when no name occurring in P is structurally equal to a name occurring in any other locus. Communication requires such an equality, so no receipt of P fires against the environment and P ’s stack receives no credits. With a positive burn rate it reaches zero in at most $\lceil \sigma_P / \varepsilon \rceil$ steps, where ε is the minimum debit. $\square$

Remark 33 (Closure is luck, not construction). The structural reading of Lemma 32 is weaker than the restriction-based one in an instructive way. Because name manufacture is decentralised (Remark 1), two computations may independently produce structurally equal names, so a locus cannot guarantee closure by construction — only observe that it currently obtains. Perfect isolation is not purchasable. We will find in §12 that the same fact, read the other way round, is what makes recombination generative.

Corollary 34 (Concealment is a luxury of the starving). *Metabolic viability lower-bounds observability: any computation that survives indefinitely must expose surface, hence must be perceivable, hence is in principle predictable and so in principle edible.*

The predator’s counter-move is now legible, and it is a reading of [2, Rem. 2] turned around. Restriction makes a name unobservable but not undescribable; $Hx.\varphi$ reaches under the binder and speaks about what is hidden. A predator therefore attacks what it cannot observe by *describing* it, which is why the spatial layer and not the modal one is the hunting instrument, and why senses had to be admitted in §3 for the game to be non-trivial at all.

9.2 The invasiveness ladder

Table 1 ordered the three grades of access by what they reveal. They are ordered in the opposite direction by what they destroy.

engagement	effect on the subject	information	nutritive?
perceive	none	surface only	no
assay	perturbs; the environment changes	behaviour under one probe	no
break in	fatal	the whole interior, as structure	yes

Table 2: Information gain and subject survival are in strict opposition, and only the fatal end of the ladder pays.

Proposition 35 (The opposition is monotone). *Along the ladder, the information yielded is non-decreasing and the subject’s residual life is non-increasing; and the only rung that credits the scientist’s stack is the last.*

This is where the note’s arc closes. A scientist that is very good at forming hypotheses locates metabolic channels efficiently; locating them efficiently is eating efficiently; eating efficiently exhausts the environment that was confirming the hypotheses. *The better the theory, the faster it destroys the conditions of its own confirmation.* Success is self-terminating, and no additional assumption was needed to make it so.

10 Mixed environments

The environment contains a mixture: bare token stacks, non-scientist mortal computations, and other scientists. Each stratum behaves differently and the third is the interesting one.

Bare stacks are the primary resource: inanimate, undefended, cheap to open, and finite. They are what makes the shallow-and-external regime of §13 a regime in which science does not pay — when food lies in the open, senses suffice and a theory is pure overhead.

Non-scientist mortal computations burn tokens and may have structure, but form no model. They defend passively, by restriction, and are subject to Lemma 32 like everyone else.

Scientists model back, and three consequences follow.

Proposition 36 (Scientists are preferred prey). *A scientist exposes more surface than a passive mortal computation, since inquiry is a dialogue of depth at least two conducted over channels that are observable across the cut, whereas metabolism alone requires only enough surface to harvest (§12.6, Proposition 63). By Corollary 34 a scientist is therefore more readily perceived and its metabolic channel more readily derived. The comparison is structural and needs no normalisation by stack size or burn rate.*

The most epistemically active agents are the most edible. We do not think this is an artefact.

Theory theft. Reflection makes another scientist’s hypothesis readable. A hypothesis in guard position is a term; a term may be quoted; and @ φ inspects the quotation without communicating. So a scientist may acquire a model by reading rather than by experimenting — cheaper, and not fatal to anyone. Whether honest science survives this depends on whether holding a model requires exposing it, and in this calculus it does, because a guard is surface. We record the equilibrium question as an open problem (§18) rather than settle it.

Imagination. Turing completeness plus reflection means a scientist may carry $\llbracket - \rrbracket$ itself as data and *run* a model of a prey’s metabolism inside its own budget, paying tokens to avoid paying tokens.

Proposition 37 (When imagination pays). *Simulating a hunt is viable exactly when the cost of the simulation is below the expected cost of the errors it prevents:*

$$\kappa_{\text{sim}} < \Pr[\text{failed hunt}] \cdot (\kappa_{\text{sense}} + \kappa_{\text{assay}} + \kappa_{\text{break}}).$$

This is the point at which the scientist stops being a reactive forager and becomes a planner, and it is where the stacking of the cost and history monads acquires an interpretation that is not merely formal: the simulated prey is itself a mortal computation, metered inside the predator’s budget.

11 Trophic structure

11.1 Conservation is the right scope, not a limitation

Smith’s organisation of the biosphere [22] distinguishes organisms that consume other organisms from those that consume only external energy, and it is tempting to read the second class as evidence that a model of life needs an open system — a source outside the conserved pool. We do not think that reading survives inspection, and we record why, because the temptation is to apologise for conservation at exactly the point where conservation is doing the work.

The sun is a reservoir of finite extent, situated in a physical system in which energy is conserved. Its apparent externality is an artefact of where the boundary was drawn: put the boundary at the biosphere and the sun is outside it; put the boundary at the solar system and there is one conserved budget being spent down. Openness, here as generally, is a bookkeeping convenience.

So conservation is not a disanalogy with Smith’s picture. It is the picture at the correct scope, and the whole trophic stratification is recoverable without minting anything.

11.2 The distinction is an access profile

If the sun is a token stack, then the autotroph and the heterotroph are not distinguished by *what* they eat. They are distinguished by how the stack they eat may be accessed.

	a source	a prey
release	rate-limited; cannot be accelerated	all at once, on demand
exhaustible within a lifetime	no	yes
exclusive	no; many may harvest at once	yes; a linear receipt
adaptive	no	yes, and it models the harvester back

Table 3: The trophic distinction is an access profile, not a kind of substance.

The calculus carries this distinction natively, and no new sort is required to express it.

Definition 38 (Source and prey). A metabolic channel is a *prey* channel when its stack is offered by a linear send, so that a single receipt exhausts it. It is a *source* channel when its stack

is offered by a persistent emitter — a replicated or contract-driven send delivering bounded quanta indefinitely — so that receipts are non-exclusive and rate-limited by the emitter rather than by the harvester.

Same sort, same rewrite rule, different multiplicity. Whether a locus is a source or a prey is therefore a property expressible in the generated logic, and a mortal computation’s diet is fixed by which of these profiles its senses are tuned to derive.

Remark 39 (Trophic type is a fact about the cut). It follows that trophic type is not intrinsic. Whether something counts as external energy or as another organism depends on which side of the cut of §2 it sits, and at what granularity one looks: a colony resolves into members, a member into a source and a consumer sharing a boundary. Since in this setting the boundary is a restriction, “external energy” has a formal referent and a relative one. Smith’s stratification is thereby recovered as a statement about how a world is cut, not as a statement about two kinds of stuff.

11.3 The split is predicted, not stipulated

The payoff is that we do not have to posit the trophic division. Two results already in hand imply it.

Remark 10 observes that a recursion following the *structure* rather than the reduction is exact at a finite unfolding. A rate-limited persistent emitter is precisely such a recursion: a replicated send, structurally regular, non-adaptive. Proposition 9 observes that an environment whose recursion follows the *reduction* admits no characteristic formula exact at any finite unfolding. An adaptive prey, being itself a mortal computation, is precisely such an environment.

Proposition 40 (The two sciences). *A mortal computation feeding on source channels faces an environment whose complete theory is exact at a finite unfolding: its inquiry converges, and after convergence its epistemic outlay falls to the rent on a fixed formula. A mortal computation feeding on prey channels faces an environment that revises against it, so no finite theory is ever complete and a permanent revision cost is incurred for as long as it lives.*

Corollary 41 (Intelligence is a heterotroph’s expense). *The standing epistemic cost of a strategy is bounded for source-feeders and unbounded for prey-feeders. Hence the apparatus of continuing inquiry — sensing at depth, holding and revising hypotheses, simulating before acting — is worth its rent only to the second.*

Plants do not need brains because the sun is the environment whose complete theory is affordable. We offer this as a consequence of Proposition 9 and Remark 10 rather than as an observation imported from biology; that it is also true of the Earth we take as corroboration of the modelling rather than as its source.

11.4 Why the overlap is thin

The space of mortal computations is plainly large enough to contain those that eat both — nothing in Definition 38 forbids tuning one’s senses to both profiles. On the Earth the corresponding region is sparsely occupied, and it is worth asking what the formalism says about why. Four pressures are available, none of them requiring new apparatus.

1. **Opposed geometry.** Yield from a rate-limited dilute source scales with collecting surface; yield from a concentrated one-shot source scales with reach and speed. Surface against velocity, derived from Table 3 alone rather than borrowed from the shapes of leaves and predators.
2. **Opposite sides of Proposition 9.** By Proposition 40 the two diets demand different points on the adequacy/checkability trade-off. The source-feeder can live in a terminating fragment; the prey-feeder cannot, since its subject matter is Turing complete and adaptive. One budget cannot be allocated to both regimes at once.
3. **Double rent for single-mode yield.** A foraging strategy is a hypothesis together with a sensory schedule, and both are held at a cost. A dual-mode computation pays rent on two apparatus while eating with one at a time — the argument of §16 applied to strategies rather than to formulae.
4. **Hysteresis.** Revision is a walk on the relaxation lattice and walking costs. Where the transition cost between strategies exceeds the gain from switching, the mixed strategy is not infeasible but dynamically unstable: it is not a fixed point of the revision dynamics, and specialisation is an attractor.

The sharpest statement combines the first two with §9.

Proposition 42 (The mixed strategy is the worst of both). *A computation harvesting both profiles carries the prey-feeder’s permanent revision cost (Proposition 40) while maintaining the source-feeder’s large collecting surface. By Lemma 32 and Proposition 36 that surface is exactly what makes it perceivable and its metabolic channel derivable. It therefore bears the heterotroph’s epistemic outlay together with the autotroph’s vulnerability.*

The region is thin because it is dominated, not because it is unreachable.

11.5 The overlap is populated by composites

There is a further and we think better answer, which is that the question contains a false premise.

On the Earth, dual capability is realised almost entirely by symbiosis: corals with their zooxanthellae, lichens, kleptoplasty in *Elysia*, and — decisively — the mitochondrion and the chloroplast themselves. Very few single genomes do both at scale; pairs of genomes sharing a boundary do it constantly.

In this setting that is

$$\text{new } m \text{ in } (P \mid Q),$$

two specialists sharing a metabolic channel under a restriction. Parallel composition makes composites first-class and graded separation $|_k$ measures how much interface they share, so the construction needs nothing added.

Proposition 43 (Composite resolution). *The overlap region is not underpopulated; it is populated by terms that are parallel compositions rather than atoms. The restriction permits each component to specialise its access profile — one tuned to sources, one to prey — while the shared metabolic channel pools the yield, so Proposition 42 does not apply to the composite.*

Read with Remark 39 this is not a special construction but the general situation: the organism boundary is drawn by the restriction, not by the strategy, and what looks like one dual-feeding thing at one granularity is two specialists at another.

11.6 Succession

One consequence of taking conservation as the correct scope, rather than as a limitation to be repaired, deserves recording. Because a source is finite — rate-limited, but not inexhaustible — a world does not settle into stable coexisting trophic layers. It runs through them.

Proposition 44 (Trophic succession). *While sources deliver at a rate satisfying Proposition 27, source-feeding dominates and the cheap fragment of §5 suffices. As sources deplete, the foraging inequality fails for them before it fails for prey, since prey concentrate what sources have already collected. Prey-feeding, and with it the permanent revision cost of Proposition 40, is therefore forced rather than chosen. Thereafter κ accumulates until the inequality fails for every locus.*

The stratification is temporal as much as structural. On a stellar timescale this is a distant prospect and not a modelling artefact; it is the correct statement of the situation at the scope at which energy is conserved.

11.7 An ensemble test

Both claims of this section — that the mixed strategy is dominated, and that the composite is not — are checkable inside the framework rather than merely analogical, and the machinery is that of §13. Sample ecologies across the (ι, δ) plane and ask, first, whether the mixed access profile is ever a viability optimum for an atomic computation, and second, whether the composite of Proposition 43 dominates it wherever it is not. We regard this as a sharper first computation than calibrating the phase boundaries, because it can come out either way and one would want to know which.

12 Reproduction

Harvest ends a computational life, so an ecology with no births is a monotone depletion of agents and nothing in it persists long enough to be a subject matter. This section supplies reproduction. The mechanism is reflection, and the interest of it is that the genetic apparatus is not modelled but found: a genome is a quotation, a locus is a context, and crossover is the exchange of parands between two quoted terms.

12.1 The naive route, and why sex is worth its price

The cheap route is already in the calculus. Replication, guarded recursion, a persistent receipt, a contract: each reinstates a term exactly. This is asexual reproduction, and it is strictly cheaper than the alternative — no partner, no alignment, no disclosure, no recombination to compute. Sexual reproduction produces variants rather than copies, so the two are not different apparatus but the extremes of a single dial: exact reinstatement at one end, recombination at the other. In the sexual regime there is no exact recursion anywhere in a lineage.

Why pay for the dial to be off zero? The usual answers are available, but this setting supplies a sharper one, and it comes from the first half of the note.

Proposition 45 (Sex attacks the predator’s hypothesis language). *By Definition 25 a predator must derive its prey’s metabolic name, and by Corollary 30 the hypothesis licensing that derivation is a description of a namespace. Recombination at homologous contexts alters which channels a lineage’s members carry. Hence sexual reproduction perturbs precisely the object about*

which the predator’s hypothesis is formed, whereas variation confined to continuations perturbs only behaviour under a hypothesis that remains correctly aimed.

This is a Red Queen argument with a mechanism rather than an analogy, and it is available because the hypothesis language was chosen to be a logic of namespaces in the first place.

12.2 The germ is a quotation

Definition 46 (Genome, phenotype). The *genome* of a mortal computation whose code is P is the name $@P$. Its *phenotype* is $*@P$.

Everything one wants from the distinction follows from properties the calculus already has.

- $@P$ does not reduce. A quotation is a name, and names do not step. So a genome does not metabolise, does not age, and burns no tokens while held.
- $@P$ is transmissible. Names are what sends carry, so a genome may be handed to a partner or to an offspring on an ordinary channel.
- $@P$ is inspectable. By §3 a name predicate sees into a quotation, and by Remark 1 it sees all the way down. A genome may therefore be screened without being run — at reflection prices, which by Table 1 are the cheapest available.
- $*@P$ costs. Running is what is metered.

Remark 47 (What we do and do not claim here). It is tempting to read the first item as a formal Weismann barrier. We decline. What the calculus gives is an asymmetry of *cost* — held quotations are free, running terms are metered — and that is all we use. Whether an individual’s acquired revisions can be written back into what it transmits is a separate question, on which the empirical case against a strict barrier is considerable [21], and which is not settled by any theorem here. We note only that a scientist’s hypotheses are held as data rather than as code, which puts them on the near side of any such barrier, and we leave the matter as Open Problem 1.

Remark 48 (Persistence and reproduction are distinguished by the ledger). Syntactically, a term that reinstates itself and a term that spawns a copy can look alike. Under conservation they are not alike: a reinstatement that continues to draw on the parent’s metabolic channel is soma, and one that must be endowed from it is offspring. The germ/soma boundary is drawn by provisioning, not by grammar.

12.3 The normal form is a tree, and a locus is a context

Up to structural congruence a process is a parallel composition of primes, and each prime is a receive, a send, or a drop:

$$P \equiv \prod_{i \in I} \text{for}(p_i \leftarrow c_i) P_i \mid \prod_{j \in J} c_j!(Q_j) \mid \prod_{k \in K} *x_k.$$

This is a gross decomposition only. Each continuation P_i and each transmitted body Q_j is itself a process with a normal form of the same shape, so the decomposition recurses and a genome is a finitely branching *tree* whose nodes are parallel compositions and whose edges are prefixes. The parands are the individual primes, at every level.

Definition 49 (Locus, allele). A *locus* of P is a one-hole context $K[-]$ obtained by deleting a parand from the tree of P ; the deleted parand is the *allele* at that locus. A locus is thus identified by the path of channels from the root to the hole, not by a channel alone.

The vocabulary is not new: these are the contexts of the context-labelled modality $\langle K \rangle$ of §3. Genetics requires no notion of address the note did not already have.

Definition 50 (Homology). Loci K_M of the mother and K_F of the father are *homologous* when their channel paths from the root agree and their binding environments agree — that is, the prefixes along the two paths bind the same pattern variables.

Homology is therefore a partial isomorphism of the two trees rather than a matching of two multisets, and the binding clause is a reading-frame condition: an allele’s body may reference only variables its new context binds.

Proposition 51 (Species is a namespace formula). *Two computations can recombine only at their homologous loci, so the extent to which they interbreed is the extent to which their path sets agree. Reproductive isolation is namespace disjointness, and conspecificity is expressible as a formula over contexts in the generated logic.*

12.4 Crossover

Definition 52 (Crossover). Let H be a set of homologous loci of $@M$ and $@F$ and let $\chi : H \rightarrow \{M, F\}$ be a choice function. The *recombinant* $@C$ is the term obtained by filling each locus in H with the allele of the parent χ selects, retaining polarity — a receive is replaced by a receive, a send by a send. Where the split is taken between a guard and its body, Definition 50’s binding clause is required and is what keeps the recombinant well formed.

We take crossover proper rather than assortment of whole primes. Assortment — exchanging a prefix together with its body — is the safe special case in which the binding condition is automatic; it is Mendel’s independent assortment, and it cannot recombine within a gene. Splitting guard from body is where recombination earns its name.

Proposition 53 (Effect size is graded by depth). *A locus at depth 0 exchanges an entire subsystem; a locus at depth d exchanges detail interior to a behavioural sequence of length d . Hence the perturbation induced by a swap is non-increasing in the depth of its locus.*

The gradation is uniform, and it is your own observation about recursion that makes it so: were the genome to contain recursive definitions, a swap inside a recursive body would be a global edit whose effect size was large regardless of depth. Since sexual reproduction yields variants and never exact reinstatement, no such loci occur, and the analogue of the distinction between large-effect regulatory and small-effect structural variation falls out of the term structure rather than being posited as a rate.

12.5 Communicating pairs as units of inheritance

Definition 52 lets a choice function select any allele at any homologous locus, and §12.8 then observes that most of what it produces is inviable. There is a discipline which removes much of that waste, and it changes what the unit of heredity is.

Definition 54 (Interaction graph, communicating pair). The *interaction graph* of a genome has the primes as nodes and an edge between a send and a receive whenever they act on the same channel with compatible pattern. A *communicating pair* is such an edge together with its two endpoints. The criterion is syntactic by necessity: whether two primes *will* communicate is a reachability question, and settling it is not something a parent could afford (Proposition 70).

The unit of heredity is a redex. A gene is not a piece of structure but an interaction.

That is worth pausing on, because it obtains something one would otherwise have to impose. Material that must work together is inherited together — linkage — not because a chromosome places it adjacently but because the pairing *is* the working-together.

The discipline does not by itself preserve viability, and the reason is instructive. Suppose the mother contains a race,

$$c!(P) \mid \text{for}(x \leftarrow c)Q \mid \text{for}(y \leftarrow c)R,$$

and the swap removes the pair $(c!(P), \text{for}(x \leftarrow c)Q)$ in favour of a father’s pair on some other channel. The donated interaction is sound; the *residual* receipt on c is orphaned, which is precisely the deadlock half of the balance condition. Removing a pair cannot break the material it donates but can break the material it leaves behind.

Proposition 55 (Balance is preserved exactly by degree-preserving swaps). *Let a swap remove a communicating pair on channel c and install one on channel c' . If $c = c'$, the number of sends and the number of receipts on every channel are unchanged, no prime is orphaned, and any race on c survives with the same arity. If $c \neq c'$, the residual primes on c may be orphaned and the balance condition of §12.8 may fail.*

So preserving races is not a second benefit alongside preserving balance; it is the side condition under which the balance guarantee holds at all.

Remark 56 (Discipline substitutes for screening). Proposition 70 says a parent cannot screen its recombinants. Under a degree-preserving pair discipline it does not have to screen for balance, because the recombination operator cannot produce an unbalanced offspring. The cost is paid once, in the machinery that constrains the operator, rather than per offspring in model checking. This is an argument for why recombination apparatus should be expected to be *constrained* rather than free, and it depends on Definition 54’s criterion being syntactic — a semantic notion of “will communicate” would hand the saving straight back.

Remark 57 (A dial, and it is the dial of §6). Three disciplines are now in view: free parand swap, maximal in variation and mostly lethal; pair swap, which guarantees the donated interaction but risks the residue; and degree-preserving pair swap, which guarantees both and varies least. Each rung buys viability with explorable distance — which is the trade-off of Propositions 12 and 14 seen at the scale of a lineage. It follows that §6.5’s claim is conditional: recombination escapes confinement only if the discipline is loose enough to make a shallow re-branch, and looseness is paid for in dead offspring. Where the optimum lies is an ecological question, and belongs with the ensemble of §13.

Remark 58 (What the discipline does not simplify). Pairs overlap. A send may match several receipts, a receipt’s continuation contains further sends matching elsewhere, and a persistent

receipt matches many sends over time, so the interaction graph is not a matching. “Swap a pair” therefore means “swap an edge together with its endpoints, where the endpoints carry other edges,” and Proposition 55 is a statement about one channel at a time. The multi-channel case requires the degrees preserved simultaneously, which is a stronger and rarer condition, and we do not claim it is generic.

12.6 Three kinds of channel, and what a swap can touch

The discipline of §12.5 operates on some of a genome and not on all of it, and saying which requires a classification that is useful well beyond recombination.

Definition 59 (Channel types). Let $W \equiv P \mid E$. A channel c occurring in P is

internal if P contains a matching send and receipt on c and no name structurally equal to c occurs in E : only τ steps arise from it;

external if a name structurally equal to c occurs in E but P contains no matching pair on c : its steps are interactions and nothing else;

both if P contains a matching pair on c and a structurally equal name occurs in E .

The classification is relative to the cut, as trophic type was (Remark 39), and for the same reason: what counts as inside is a question about where the boundary was drawn.

Remark 60 (This is the surface, made syntactic). External and both-type channels are exactly the context-observable features of a computation. They are what a structural predicate evaluated at the surface can reach (§3), what a context-labelled modality can exercise, and what Lemma 32 was about: a computation all of whose channels are internal has no surface, cannot forage, and starves. The three grades of access in Table 1 and the three channel types are the same distinction seen from the two sides of the cut.

Dialogue

Definition 61 (Dialogue, and its depth). A *dialogue* is an alternating chain: a step on an external or both-type channel, a causally connected sequence of internal reductions, and a further step on an external or both-type channel whose availability depends on the first. Its *depth* is the number of alternations.

A computation with no dialogue of depth greater than one is a reflex: whatever it emits is not conditioned on anything but the message immediately received. Depth two and above is conversation — the environment’s reply is answered in the light of what the first exchange produced. Two consequences follow, and the first is what the user of this framework will care about.

Proposition 62 (Conversation is bounded by the stack). *Each alternation of a dialogue debits at least the internal chain connecting its two external steps. Hence the depth of dialogue a computation can sustain is bounded by its token supply, and by Proposition 6 the modal depth at which it can be observed to differ from a rival is bounded by the same quantity.*

Proposition 63 (Inquiry has a structural signature). *A computation capable of inquiry has a dialogue of depth at least two on both-type channels: the assay of Definition 4 emits a probe, receives, conditions on what it received, and emits again. A computation whose both-type channels support only depth-one dialogue can forage but cannot revise. Since dialogue depth is a nesting of context-labelled modalities over channels the classification identifies, “this computation is a scientist” is a formula of the generated logic rather than a designation we assign.*

That repairs something the note has so far taken on trust. The grazer of Appendix A is not a non-scientist because we declined to give it a hypothesis; it is a non-scientist because its interaction graph contains no depth-two loop through a both-type channel, and the difference is checkable.

It also repairs Proposition 36, whose original statement had to normalise by burn rate. The correct statement is structural: deeper dialogue requires more both-type channels, and both-type channels are surface.

What each kind of swap touches

swap	balance	observability	role
internal pair	preserved	silent: τ only	cryptic variation, priced but unseen
both-type pair	preserved if degree-preserving	alters the dialogue	adaptive variation
external singleton	internally unaffected	alters the interface	a wager on the environment

Table 4: The classification says which discipline applies where. Only internal and both-type channels host pairs at all; external material consists of primes whose counterpart lies in the environment, so it is swappable singly without orphaning anything — at the price that its viability is a bet about the environment rather than a fact about the genome.

This grading is independent of the depth grading of Proposition 53. A recombination policy has two dials: how much structure a swap moves, and how observable the moved structure is.

Proposition 64 (No neutral variation). *A swap of an internal pair is invisible to weak bisimulation, hence to every context, hence to any predator’s hypothesis. It is not invisible to the ledger: its reductions are metered like any others. There is therefore no behaviourally silent variation that is also metabolically silent, and selection acts on internal variation through efficiency rather than through function.*

Corollary 65 (Cryptic variation and its release). *Internal variation accumulates without behavioural consequence and becomes consequential exactly when its channel is reclassified — which happens when a structurally equal name comes to occur across the cut. By Proposition 68 recombination can do this, since a transplanted pair may carry a channel that coincides with a resident external name. A lineage may therefore accumulate variation cheaply while remaining illegible, and expose it later.*

This sharpens Proposition 45. Only the observable fraction of recombination attacks a predator’s hypothesis, because a predator’s hypothesis is a description of the external and both-type sub-namespaces and nothing else. But internal variation is the reservoir from which observable variation is later drawn, so the defence is two-staged: accumulate where it cannot be seen, and expose when the exposure is worth its cost.

Remark 66 (Dialogue is where the impact is). Collecting the section: swapping at redexes concentrates variation on channels that carry interactions, and among those, the both-type channels are the ones whose interactions are legible across the cut. A discipline that swaps redexes therefore acts, by construction, on the computation’s dialogue with its environment rather than on structure that no context could ever observe. That is the sense in which the discipline of §12.5 has real consequence: it is not merely a filter for viability but a filter that concentrates variation where it can matter.

12.7 Names, freshness, and the bound on growth

Because `new` is sugar (Remark 1), names have size, and size is priced. One might therefore fear that recombination inflates them: if a child’s fresh names were manufactured from its enclosing term, and its enclosing term combines two parents, names would double with each generation and a lineage would price itself out of existence within a few dozen.

The calculus forbids it.

Proposition 67 (Growth is slaved to term size). *No process contains a name that is its own code, since quotation strictly increases size. Hence every name occurring in a process is strictly smaller than that process, and no admissible desugaring manufactures a name from more than a proper part of its scope. Name size cannot drive term growth; it is bounded by it.*

The feared scheme is not merely expensive but syntactically impossible, and the bound holds uniformly across every decentralised scheme — which is why we may remain agnostic about which one is taken.

What growth remains is the right kind. At a homologous locus the recombinant takes one allele, not both, so aligned material contributes a pointwise maximum rather than a sum. Growth enters only at *non-homologous* loci — material present in one parent alone — where inheriting the union is additive per generation. That is gene duplication, it is linear, and by Definition 3 it is priced, which is the regime in which selection can act on it.

Proposition 68 (Decentralisation makes recombination generative). *Since names are manufactured locally and freshness is scope-relative, two parents may independently hold structurally equal names at loci private to each. A recombinant inheriting both then possesses a single shared channel where its parents had two private ones. Recombination can therefore create interaction paths neither parent had, rather than only permuting existing ones.*

Under a central name server, or under a calculus of atomic names with global freshness, this route is closed. It is the same fact recorded in Remark 33: what denies a locus guaranteed closure is what allows a lineage novelty.

Remark 69 (Accumulation, and a reading of senescence). Names grow because the terms containing them grow, and a scientist’s term grows as it accumulates probe channels, held hypotheses, and machinery. With sensing priced by depth, every operation then costs more, and a computation that does not reclaim dead structure eventually cannot afford to run. This is a discipline-dependent claim rather than a theorem — garbage collection is the alternative, and is itself priced — but under the discipline it says something worth saying: an organism ages because it accumulates, and accumulation is what learning is.

A condensation pass at birth, canonically renaming the recombinant’s private names from a minimal seed, resets the inherited weight at a cost proportional to genome size. We note that

this is an argument about *cost* and is orthogonal to Remark 47: whatever the traffic between soma and germ turns out to be, the weight must be renormalised or lineages accumulate without bound.

12.8 Viability, and why there is selection rather than design

A parallel composition of primes is always a term, so syntactic well-formedness is free. Viability is not, and it is a conjunction of conditions the generated logic can state: closure, the reading-frame condition of Definition 50, balance — receives without matching sends deadlock, sends without receipts leak — and liveness in the marked-graph sense. Screening a recombinant is therefore model checking, and the classifier apparatus of [2] is the embryology of this note.

But model checking is metered, and the space is large.

Proposition 70 (Selection is post hoc because screening is priced). *Aligned genomes with n homologous loci admit 2^n recombinants, where n is the size of the genome and not the width of its top level. Screening each is an outlay against the same stack that funds foraging and inquiry. Hence a parent cannot emit only viable offspring — not for want of a criterion, but because applying it exhaustively is unaffordable.*

Reproduction is a gamble because screening is priced. Were model checking free, a parent would design its offspring and selection would have nothing left to do.

Neither fine granularity nor cost accounting gives this alone; it needs both.

12.9 The energy tension: a three-way allocation

Reproduction takes energy, and under conservation the energy comes from the parents and nowhere else. An offspring is endowed out of a parent’s stack, so every child is a portion of the parent’s remaining life.

Proposition 71 (Minimum viable endowment). *An offspring endowed below the outlay required for its first successful harvest — $\kappa_{\text{sense}} + \kappa_{\text{assay}} + \kappa_{\text{break}}$ for the shallowest prey available to it, by Proposition 27 — dies before it eats. Provisioning is therefore bounded below, and the number of offspring a parent can endow is bounded above by its stack divided by that floor.*

We take the endowment to be chosen by the parent rather than fixed by the offspring’s code size, which makes the trade-off between many poorly provisioned offspring and few well provisioned ones a strategy rather than a constant.

The tension the reader will have anticipated is not two-way but three-way. One conserved stack funds soma maintenance, germ production, and epistemic outlay: the same tokens buy eating, breeding, and thinking.

Corollary 72 (Inquiry trades against fecundity). *By Corollary 41 the standing epistemic outlay is unbounded for a prey-feeder. By Proposition 71 offspring are bounded below in cost. Since both draw on one stack, a strategy that sustains deep inquiry sustains fewer offspring.*

This supersedes the reading of surplus offered in §16: surplus does not simply buy curiosity, it buys curiosity *or* children, and which it buys is the life-history problem.

12.10 What reproduction does for the science

Two consequences bear directly on the first half of the note.

The lineage is a second learner. A scientist’s hypotheses are guards; guards are parands; parands are what crossover exchanges. So the genome contains the hypotheses, and they are revised by two mechanisms on two timescales — within a life by walking the relaxation lattice of §5, across lives by recombination. The same object is revised by both. A lineage is not merely a means of persistence; it is doing science by other means, and the proximate/ultimate distinction of §16 extends to cover it.

The correspondence is tighter than an analogy. A revision move is a locus in the formula tree together with an operation there (Definition 13); a crossover is a locus in the term tree together with a choice of allele (Definition 52). The characteristic-formula construction of §5 carries term structure to formula structure, so a swap at a term-context K induces a revision at the formula-locus K corresponds to. The two mechanisms are one move, transported along χ , and applied at two timescales. What distinguishes them is the accessible distance: by Proposition 12 an individual’s incremental walk cannot leave its ball, whereas a swap high in the term tree is a shallow re-branch by construction. *Recombination is how a lineage makes the displacements its members cannot.*

Reproduction is what makes induction pay. Corollary 30 forced hypotheses to be about namespaces, since destructive assay leaves an individual-level hypothesis worthless after its own first test. But a namespace with no live members is worthless too. Reproduction repopulates the namespace, so a class-level hypothesis retains value across the specimens it costs to test.

Proposition 73 (Kinds are the condition of inquiry). *In an ecology without reproduction the scientist monotonically destroys its own subject matter and no hypothesis retains extension. With reproduction the population of a namespace is sustained while the foraging inequality holds for its members, and hypotheses about kinds remain true of live instances. Species are what make science possible.*

Remark 74 (What reproduction does not do). It does not repeal conservation. Tokens are redistributed into new loci and never created, so Proposition 44 stands unaltered and the world still runs down. What changes is the shape of the decline, not its direction: births convert a monotone extinction of agents into a population dynamics with a carrying capacity, which is what makes the intermediate band of Figure 2 occupiable for longer than a single generation.

13 Distributions over ecologies

13.1 A world is a partition of a conserved quantity

Because Θ is fixed and never minted, a world is a partition of Θ across loci together with a structural assignment saying which loci are scientists, which are non-scientist mortal computations, and which are bare stacks. The configuration space is graded by Θ , and a distribution over it is a microcanonical ensemble in the strict sense — fixed total, distribution over microstates. The analogy is exact rather than suggestive, and it is exact because conservation is exact.

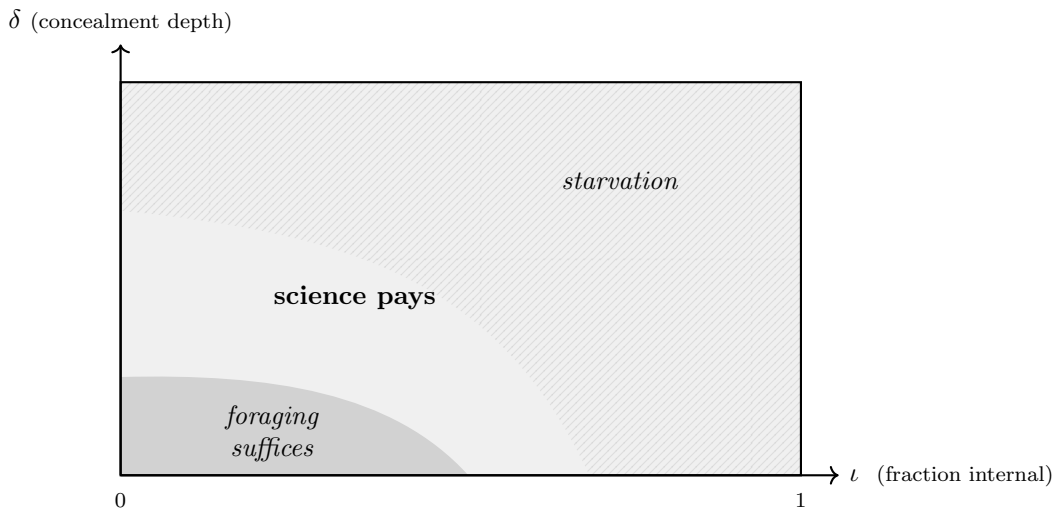


Figure 2: The phase diagram. When food lies shallow and in the open (low δ), senses alone suffice and a theory is overhead: the pure forager beats the scientist. When too much of Θ is already internal, or lies too deep to be found for what it yields, discovery costs more than it returns and everything starves. Hypothesis formation pays in the intervening band. Science is a phase, not a universal advantage.

Definition 75 (Configuration, ecology). A *configuration* is a finite multiset of mortal computations and bare stacks together with an assignment of stack sizes summing to Θ and a designation of which computations carry a revision policy. An *ecology* is a configuration together with a distribution over the configurations reachable from it.

13.2 Two order parameters, and a phase

Two quantities govern the qualitative behaviour and they are independent:

- $\iota \in [0, 1]$, the fraction of Θ already *internal* to scientists rather than loose in the environment;
- $\delta \geq 0$, the mean *concealment depth* of external stacks — how far under restriction and structure they lie, which by Definition 3 sets the sensing cost of finding them.

Science only pays in an intermediate band — where food is neither free nor absent.

We regard this as the most substantive empirical claim the construction makes, and it is a claim the construction makes rather than one we impose: the boundaries of Figure 2 are where the foraging inequality of Proposition 27 changes sign. The low- δ boundary is where κ_{sense} falls below the cost of holding a theory at all; the upper boundary is where Proposition 27 fails for every available P .

13.3 Sampling

The machinery for turning a configuration into a distribution and sampling trajectories through it is not new work. The construction of [7] takes a decorated continued interactive GSLT, an

adequate spatial–behavioural logic, and a weight-map decoration to a Gillespie-style simulator, framed as a functor from a Grothendieck construction of decorations into weighted coalgebras. That is precisely what an ecology is: the weight map turns a configuration into a distribution, and the Gillespie construction samples it. §15 takes up what that fibred structure means here.

14 The audit: what the assumptions already fix

It is easy to build a machine that fits a story. We therefore separate, explicitly, what follows from the commitments and what we chose.

14.1 Forced

feature	forced by
the hypothesis language	OSLF applied to the rho presentation; not designed
formulae denote bisimulation classes	adequacy of the generated logic
exactly three grades of access	the calculus offers no fourth (§2)
senses are structural predicates	admitting the structural layer, priced by depth
an assay is a guarded receipt	the where clause
refutation is budget-relative	Proposition 6
the hypothesis space	the relaxation lattice of [2]
hypotheses must be about namespaces	destructive assay (Corollary 30)
eating is an ordinary receipt	internalisation plus Turing completeness
death is failure to fund the next step	the cost monad
$\sigma + \kappa = \sigma_0$ globally	the history monad plus no minting
predators are non-image contexts	Theorem 24
the exposure lemma	Lemma 32
optimal foraging, and its profitable band	conservation plus depth pricing (Prop. 27)
information opposes survival	Table 2
a genome is inert, transmissible, inspectable	quotation does not reduce (Def. 46)
a locus is a context	the recursive normal form (Def. 49)
name growth is bounded by term size	no term contains its own code (Prop. 67)
selection is post hoc	screening is priced (Prop. 70)

14.2 Chosen

The physics is forced. The ecology and the psychology are the free parameters.

Everything metabolic and everything epistemic follows from cost-accounted rho together with the logic it generates. What remains to be chosen is the weight map, the initial distribution, and how the scientist thinks. We regard the smallness of that list as the principal evidence that the modelling choices were not tuned to the conclusions.

parameter	status
Θ , the conserved total	free; sets the scale of the whole game
the initial configuration and its distribution	free; §13
the admitted context class $\mathcal{A}$	free; Definition 25, and it is the rulebook
the pricing schedule κ_{sense} , per-rewrite debits	free; the weight map
concealment depth δ of stacks	free; an order parameter
harvest all-or-nothing versus partial	free; we take all-or-nothing
the revision policy	free; this is the agent design, and we leave it open
the desugaring of new	free, subject to decentralisation (Rem. 1)
provisioning per offspring	free; we make it parent-chosen, so r/K is a strategy
the crossover choice function χ	free; the recombination rate in disguise

15 Is the assignment functorial?

It is natural to ask whether ecologies are a functor of the theory: is there a functor from the subcategory of cost-accounted graph-structured lambda theories to distributions over computational ecologies of the kind described above? The answer is instructive, and it is no in a way that tells us what the right question was.

15.1 Not from the base

Write $\mathbf{GSLT}_{\mathcal{C}}$ for the subcategory of cost-accounted GSLTs. The assignment $\mathcal{C}(G) \mapsto$ “its ecologies” is not determined by $\mathcal{C}(G)$. Two different weight maps — two different pricing schedules on the same cost-accounted theory — give different sensing costs, hence different foraging inequalities, hence different phase boundaries in Figure 2 and different distributions over trajectories. The object does not determine the image.

Proposition 76 (A fibration, not a functor). *The ecologies form a fibration $\pi : \mathbf{Eco} \rightarrow \mathbf{GSLT}_{\mathcal{C}}$ whose fibre over $\mathcal{C}(G)$ is the class of ecologies presentable in $\mathcal{C}(G)$, indexed by the decorations $\text{Dec}(G)$. The assignment of distributions is a functor out of the total space — equivalently, out of the Grothendieck construction $\int_{\mathbf{GSLT}_{\mathcal{C}}} \text{Dec}$ — and not out of the base.*

This is [7]’s theorem with a different label on the target: that construction already goes from a Grothendieck construction of decorations into weighted coalgebras, and an ecology-with-a-distribution is a weighted coalgebra.

15.2 A canonical section

There is nevertheless a rescue, and its shape is the interesting part.

Proposition 77 (Cost is already a weighting). *A cost-accounted GSLT is not decoration-free: the cost function is itself a weight map. Taking costs as the weights selects a distinguished decoration for every object of $\mathbf{GSLT}_{\mathcal{C}}$, hence a section $s : \mathbf{GSLT}_{\mathcal{C}} \rightarrow \mathbf{Eco}$ of π , and hence a genuine assignment of ecologies to objects.*

So the honest statement is not “there is a functor” but “there is a canonical one, and its canonicity is exactly the observation that cost is already a weighting.” Any other ecology over the same theory is a re-weighting, and the deviation from the canonical section measures how far the modeller’s pricing departs from the theory’s own.

15.3 The morphisms must be logic-preserving

Functoriality on morphisms is where a real correction is needed, and we think it is the most useful thing this section contains.

The morphisms of **GSLT** are bisimulation-preserving maps, where the bisimulation is the context-labelled one and is a congruence [3]. But an ecology depends on *spatial* structure: concealment depth, which computations sit inside which, how a network separates when a locus is removed. And the spatial layer strictly refines bisimulation — this is the entire thesis of [2], that separation says things no modal logic can say. So a bisimulation-preserving map may scramble the spatial layer, changing sensing costs and foraging inequalities while preserving behaviour exactly.

Proposition 78 (Refinement of the morphism class). *The assignment of Proposition 77 is not functorial on **GSLT**-morphisms. It requires morphisms preserving the generated logic — both the behavioural and the structural layers — which is a strictly narrower class, strictly because the structural layer strictly refines context-labelled bisimulation.*

The narrowing is not ad hoc. It is forced by the fact that the senses are structural: a morphism that does not preserve what the senses see does not preserve the ecology, because it does not preserve what it costs to find food.

Conjecture 79 (Laxity). *Even over logic-preserving morphisms the induced map on distributions is only lax. An encoding may add rewrite steps and so inflate costs, so the pushforward distorts; the natural target is distributions ordered by stochastic dominance rather than distributions on the nose.*

We state this and do not chase it. The three-line summary for the reader in a hurry is: a fibration, a canonical section via cost-as-weight, and a refinement of the morphism class forced by the senses being structural.

16 Proximate and ultimate scores

The original motivation for this construction measured the delta between hypothesis revisions and punished degradation. The game supersedes that mechanism as the *ultimate* criterion — survival is the score, and it needs no adjudicator, no referee holding a mint, and no exogenous reward signal. Two things fall out that we would otherwise have had to impose.

- **Vacuity is punished by hunger, not by rule.** The hypothesis $\top$ predicts nothing, locates no metabolic channel, yields nothing, and starves. No informativeness term is needed in any score function, because there is no score function.
- **Occam’s razor is metabolic.** A hypothesis is worth holding exactly when expected yield exceeds $\kappa_{\text{sense}} + \kappa_{\text{assay}} + \kappa_{\text{break}}$ plus the cost of holding the formula itself. Description length and food are denominated in the same currency, so minimum description length is derived rather than stipulated.

The revision delta nevertheless survives, demoted and better placed. A scientist cannot wait until it dies to learn which way to walk the relaxation lattice, so it needs a local signal, and $d(\varphi_n, \varphi_{n+1})$ together with the confirm/refute counts of §4 is exactly one. This is the proximate/ultimate distinction of behavioural ecology: viability is the ultimate criterion, the revision delta is the proximate mechanism that tracks it, and neither is redundant.

Remark 80 (The pragmatist’s objection, and surplus). It should be conceded plainly that this scientist is not a disinterested seeker of truth. Its epistemology is metabolic, and truths that do not pay go unfunded. That is the standard objection to pragmatism and we do not think it can be argued away. The construction does, however, have an unusually concrete answer: *surplus*. A scientist holding a stack larger than its foraging horizon requires can afford inquiry that does not pay, and what it buys with the excess is curiosity — though by Corollary 72 it might have bought offspring instead, and the choice between them is the life-history problem. Basic research is a metabolic luxury, available exactly to the well-fed, and in a world of conserved tokens it is therefore rare and worth protecting. We note this and leave the moral to others.

17 Conclusion: an embodied architecture, and the effectiveness of reason

We close with a speculation that is not a design. Suppose a perceptual front end — a transformer network, wired to the world, doing what such networks are unreasonably good at — were used to encode sensory data as a population of computations and token stacks of the kind this note describes. We are not proposing an interface, and nothing below depends on how the encoding is done. We are asking what would follow if one existed, because we think the consequences bear on a question older than any of the machinery here.

17.1 A division of labour

The scientist of this note lives among other computations. The world does not present itself that way, or at least does not present itself that way to us, so something must do the type conversion. That is the front end’s whole job in the proposed arrangement: not to learn, but to render the environment as an environment of the right kind — loci with surfaces, stacks, and channels an interaction can address.

The division is worth stating sharply, because it inverts the usual allocation of labour:

The network is the transducer. The ecology is the epistemology.

Nothing in §§3–12 is a learning rule in the gradient sense, and nothing in a perceptual encoder is a hypothesis. Keeping them apart is what makes the arrangement legible.

17.2 Why continual learning falls to the ecology

The features that make an architecture of this shape suited to continual learning are not additions to it. They are the results already proved.

Revision is local. A revision move is an operation at a locus in the formula tree (Definition 13), not a global adjustment of every parameter at once. Whatever is not at the locus is untouched, so the failure mode in which new learning overwrites old has no mechanism here. The corresponding cost is that a locus must be found, which is what §6 is about.

Exploration and exploitation are a budget, not a hyperparameter. The allocation between foraging, inquiry, and reproduction (§12.9) is a division of one conserved stack, and the trade-off is settled by what the ecology pays for rather than by a coefficient chosen in advance.

A population is required, not preferred. This is the sharpest of the three. Proposition 12 says an individual’s incremental revision cannot leave the ball it starts in — not slowly, but not at all. By Remark 57 the escape is recombination, which requires more than one learner and a lineage relating them. So an architecture built on this analysis does not have a population because populations are convenient; it has one because a single learner is provably confined.

Retirement has a criterion. Mortality is not a scheduling policy imposed from outside. A model that cannot fund its next step stops, and the ledger says when.

17.3 The unreasonable effectiveness of reason

Wigner’s question is why mathematics, developed for reasons internal to itself, turns out to describe a world that did not consult it [17]. Hamming’s answer is largely a selection effect: we notice the mathematics that fits and forget the rest [18]. The evolutionary answer is that our faculties were shaped by the world they model. Neither is comfortable, the first because the successes are too systematic and too predictive, the second because the mathematics that works best — for spectra, for spacetime, for objects in no ancestral environment — is exactly the mathematics no ancestor was selected on.

The architecture posited here suggests the question has been asked across a gap that is not there.

Wigner’s framing has mathematics on one side as a free creation, the world on the other as a brute given, and an unexplained correspondence between them. In this setting there is no such separation to explain. The hypothesis language is not chosen and then found to fit: it is generated by OSLF from the presentation of the computational substrate itself, and adequacy is the theorem that it separates exactly what interaction separates — no coarser, so nothing observable is inexpressible, and no finer, so nothing expressible is unobservable. The mathematics available to such a learner is the mathematics of what it can distinguish, because it is the language the distinguishing relation generates.

Mathematics is effective on exactly the domain to which interaction reaches, because it is the shadow that the access relation casts.

Two things follow which we think are more than restatement.

First, the transfer problem dissolves in the right direction. Adequacy is not adequacy for a particular environment; it is adequacy for interaction *as such*, since context-labelled bisimulation is the finest equivalence any interacting learner could resolve, whatever it is interacting with. That is why the mathematics carries to domains no ancestor met. It was never fitted to the ancestral environment in the first place — it was fitted to the form of access, and the form of access did not change when the subject matter did.

Second, there is a boundary, and its location is predicted rather than assumed. Effectiveness should degrade precisely where the relation between learner and phenomenon is not one of interaction across a cut — where there is no context to build, no probe to install, no reply to

condition on. That the domains which have historically resisted mathematisation have tended to be of just this kind is not proof, but it is the right shape of evidence, and it is at least a claim that could fail.

Finally, the categories themselves are accounted for rather than presupposed. Corollary 30 forces hypotheses to be about namespaces, because destructive assay leaves an individual-level hypothesis worthless after its own first test, and Proposition 73 sustains those namespaces through reproduction. So the traffic in kinds, classes and universals that mathematics and science run on is not read off the world and not stipulated by the knower. It is what a mortal, interacting learner can afford to hold. That is a Kantian conclusion with a ledger attached, and the ledger is what makes it a claim rather than a posture.

17.4 What is dictated, and what is not

If the mind is a computational apparatus, we think an architecture of roughly this shape is close to forced. Take the premises separately. If a mind is computational, it is ontologically isolated, so its environment is other computations and its access is interaction. If its access is interaction, its hypothesis language must be adequate for context-labelled bisimulation, on pain of being blind or wasteful. If hypotheses are to be tested rather than merely entertained, they must occupy guard position, so that a confirmation is a rendezvous. If the apparatus is finite, its budget bounds the fragment it can hold, and mortality supplies the motive without an external scorekeeper — of which none is available anyway, since a scorekeeper would be an oracle across the cut. And if revision is search on the resulting lattice, Proposition 12 requires a population.

Each of those is a derivation in this note rather than a design decision, which is what we mean by saying the shape is all but dictated. It should be equally clear what has not been shown. We have not shown that the mind is computational; that premise is doing all the work and we have assumed it throughout. We have not shown the world is computational either — only that a computational knower’s mathematics will fit whatever it can interact with, which is a claim about the knower and not about what is known. And we have not built the encoder, which is exactly the part where the difficulty of any actual system would be concentrated.

What the note does claim is narrower and, we hope, more durable: that once learning is modelled as a metered dialogue between isolated computations, the apparatus of the scientific method stops looking like something a mind does and starts looking like something a mind, so situated, could hardly avoid.

18 Open problems

1. **Inheritance of acquired characteristics.** Remark 47 declines to read the inertness of a quotation as a germ–soma barrier, and Noble’s case against a strict barrier [21] is a reason for caution rather than a settled verdict. The formal question is sharp: a scientist’s hypotheses are held as data and are therefore quotable and transmissible, whereas its metabolic and sensory machinery is code. Under what pricing does a lineage transmit the first, and what does the resulting two-channel inheritance do to Proposition 45?
2. **The plagiarism equilibrium.** Reading a rival’s model is cheaper than experimenting and kills nobody, but holding a model requires exposing it as a guard. Under what pricing does honest science survive theory theft, and is there a mixed equilibrium?

3. **Co-learning and non-stationarity.** When the environment is mostly scientists, the target of every hypothesis is itself revising. Convergence becomes a fixed-point question rather than a learning-theoretic one, and this is where we expect the geometry/matter corecursion to be readable as learning.
4. **Laxity.** Conjecture 79: is the induced map on distributions lax, and is stochastic dominance the right order on the target?
5. **The revision policy.** We left the agent design open on purpose. Is there a policy on the relaxation lattice that is optimal in the ecological sense — maximising expected residual life rather than predictive accuracy — and does it look anything like the policies learning theory recommends?
6. **Is the mixed diet dominated?** Propositions 42 and 43 are the two halves of a claim about trophic sparsity. Testing them in the ensemble of §13 — is the mixed access profile ever optimal for an atom, and does the composite dominate it where it is not — is the first computation we would run.
7. **Sharpening the phase boundary.** Figure 2 is drawn from the sign of a cost inequality. Making its boundaries quantitative for a specific pricing schedule, via the sampling construction of [7], is the first computation to do.

A A worked ecosystem

We give a small world containing two large external token stacks, three grazing non-scientist mortal computations, two reproductively isolated species of sexually reproducing scientist, and one asexual lineage. It is written in core rholang extended with the **where** clause of §4; stacks are integers and arithmetic is taken as given. The point of the listing is not that it is efficient but that every construction of the note appears in it as ordinary code, with no apparatus added.

A.1 Sources, metabolism, death

```
// A source (Def. 38): rate-limited by being a single contract, non-exclusive
// since anyone may call it, non-adaptive, and FINITE -- conservation is exact.
def Source(s, q, reserve) = {
  for (ret <- s) {
    match reserve >= q {
      true => { ret!(q) | Source(s, q, reserve - q) }
      false => { ret!(0) } // exhausted; cf. Prop. 44
    }
  }
}

// Reclaim - debit - reissue (Def. 20). Between the receipt and the send the
// stack is a message in flight: metabolism is a race condition (Rem. 21), and
// the metabolic channel m is exactly the vulnerability of Thm. 24.
def Live(m, burn, step) = {
  for (sigma <- m) {
    match *sigma > burn {
      true => { m!(*sigma - burn) | *step | Live(m, burn, step) }
    }
  }
}
```

```

    false => { Nil } // death is failure to fund a step
  }
}
}

```

A.2 A non-scientist

```

// A grazer holds no hypothesis and senses nothing beyond the one channel it
// was born knowing. By Prop. 40 its environment is a source, so its science
// was over before it began -- and it pays no epistemic rent at all.
def Grazer(m, src) = {
  Live(m, 1, @{
    new probe in {
      src!(*probe)
      | for (g <- probe) { for (sigma <- m) { m!(*sigma + *g) } }
    }
  })
}

```

A.3 Foraging as an assay

```

// The complementary guard pair of Def. 4, verbatim. The hypothesis is not
// consulted and then acted on; it sits in guard position, so confirmation IS
// the rendezvous. If the source is exhausted neither guard fires: that is
// the third outcome of Prop. 5, and it is not the same as refutation.
def Forage(m, sns, hyp) = {
  new probe in {
    sns!(*probe) // the probe perturbs
    | for (g <- probe where g != *hyp) { // predicted
      for (sigma <- m) { m!(*sigma + *g) }
    }
    | for (g <- probe where ~(g != *hyp)) { // refuted
      Revise(hyp) // walk the lattice (Sec. 5)
    }
  }
}

```

A.4 Birth: a quotation, then a drop

```

// The offspring's code is CONSTRUCTED as a name and sits inert on 'code'
// -- it does not reduce, does not age, burns nothing (Def. 46). Dropping it
// with * is what makes it a phenotype, and what starts the meter.
//
// The guard enforces Prop. 71: a parent may not endow below the outlay its
// child needs for a first successful harvest.
def Beget(m, alleles, prv, mate) = {
  for (sigma <- m where *sigma > *prv + floor) {
    m!(*sigma - *prv) // the endowment is subtracted
    | match *alleles {
      [s, h, p] => {

```



```

    new childM, code in {
      code!( @{ Body(childM, s, h, p, mate) } ) // the germ: inert
      | for (g <- code) { *g | childM!(*prv) } // the soma: metered
    }
  }
}
}
}
}

```

A.5 Crossover

```

// Three homologous loci are exposed as parameters: the sensing allele, the
// hypothesis allele, and the provisioning allele. This is assortment at
// three loci; crossover proper (Sec. 12.4) splits a guard from its body
// inside an allele and carries the reading-frame condition of Def. 50.
def Pick(a, b, bit, ret) = {
  match *bit { 0 => { ret!(*a) } ; 1 => { ret!(*b) } }
}

def Crossover(mother, father, chi, ret) = {
  match (*mother, *father, *chi) {
    ([sm,hm,pm], [sf,hf,pf], [b1,b2,b3]) => {
      new r1, r2, r3 in {
        Pick(sm,sf,b1,r1) | Pick(hm,hf,b2,r2) | Pick(pm,pf,b3,r3)
        | for (s <- r1 & h <- r2 & p <- r3) { ret!([*s, *h, *p]) }
      }
    }
  }
}

```

A.6 A sexual scientist, and an asexual one

```

// Mating requires DISCLOSURE: the genome goes out on a public channel, where
// anything able to read it can. Conspecific(s) is a name predicate (Sec. 3.1)
// holding of genomes whose sensing allele names the same source namespace as s.
// Nothing declares a species; reproductive isolation is namespace disjointness
// and nothing more (Prop. 51).
def Mate(m, alleles, sns, prv, mate) = {
  mate!(*alleles)
  | for (partner <- mate where partner != Conspecific(*sns)) {
    new chi, ret in {
      chi!( Coin3 ) // the choice function of Def. 52
      | Crossover(alleles, partner, chi, ret)
      | for (kid <- ret) { Beget(m, kid, prv, mate) }
    }
  }
}

// The shared developmental program. Alleles vary; this does not.
def Body(m, sns, hyp, prv, mate) = {
  Live(m, 2, @{ // scientists burn faster than grazers

```

```

        Forage(m, sns, hyp)
    | Mate(m, @[*sns, *hyp, *prv], sns, prv, mate)
})
}

// The asexual lineage: no partner, no alignment, NO DISCLOSURE, and no
// recombination to compute. Strictly cheaper -- and by Prop. 45 it presents
// a fixed namespace for anything that cares to learn it.
def Clone(m, sns, hyp, prv) = {
    Live(m, 2, @{
        Forage(m, sns, hyp)
    | Beget(m, @[*sns, *hyp, *prv], prv, @Nil)
    })
}

```

A.7 The world

```

new sun1, sun2, mate in {

    // Two large external stacks. sun1 is shallow and gives small quanta;
    // sun2 lies deeper and gives larger ones -- the two order parameters
    // of Sec. 13.2 instantiated as two loci rather than as a distribution.
    Source(sun1, 1, 60000)
    | Source(sun2, 8, 240000)

    // Three non-scientists.
    | new g1, g2, g3 in {
        g1!(50) | Grazer(g1, sun1)
        | g2!(50) | Grazer(g2, sun1)
        | g3!(80) | Grazer(g3, sun2)
    }

    // Species A: hunts sun1, cheap hypothesis, many small offspring.
    | new a1, a2 in {
        a1!(400) | Body(a1, @sun1, @{ q => q > 0 }, @20, mate)
        | a2!(400) | Body(a2, @sun1, @{ q => q > 0 }, @20, mate)
    }

    // Species B: hunts sun2, a discriminating hypothesis, few large offspring.
    // A and B share the mating channel and never interbreed.
    | new b1, b2 in {
        b1!(900) | Body(b1, @sun2, @{ q => q > 4 }, @150, mate)
        | b2!(900) | Body(b2, @sun2, @{ q => q > 4 }, @150, mate)
    }

    // The asexual lineage, competing with species A for sun1.
    | new c1 in { c1!(500) | Clone(c1, @sun1, @{ q => q > 0 }, @60) }
}

```

A.8 What to watch

- **Conservation is checkable by inspection.** Θ is the sum of the two reserves and the eight initial stacks. Nothing in the listing mints; **Source** only ever moves reserve into a consumer, and **Beget** only ever moves a parent’s stack into a child’s. Every debit taken by **Live** is a token leaving σ for κ , so $\sigma + \kappa = \Theta$ throughout.
- **The three outcomes are structural, not bookkeeping.** In **Forage**, refutation is an event because $\neg\varphi$ is a guard; the case where the source is exhausted produces neither event. Proposition 5 is visible as three code paths, one of which is empty.
- **Nothing screens.** No embryo is model-checked anywhere in the listing. A parent could in principle check its recombinant against the viability formula before dropping it, and does not, because the check is priced against the same stack — Proposition 70 as a programming decision rather than an argument.
- **Speciation costs one guard.** A and B share **mate** and are nevertheless isolated. Delete the **where** clause from **Mate** and they hybridise; the offspring inherits a sensing allele naming one source and a hypothesis allele tuned to the other, and starves without ever being refuted. That is the reading-frame condition failing at the level of the ecology.
- **The race is real.** **Live** and **Forage** both take from $\mathfrak{m}$ and put back. They conserve the stack and cannot deadlock, but they interleave — which is Remark 21’s point that living means racing others to your own stack, with the others here being your own body.
- **This world is at the autotrophic end.** Every organism feeds on a source, so by Proposition 40 every lineage’s inquiry terminates and none pays an unbounded epistemic rent. Note how close the other end is: species B ’s sensing allele already names a deeper namespace, and a single reallocation of it toward a grazer’s $\mathfrak{m}$ converts the lineage to heterotrophy, with Corollary 41’s standing cost arriving in the same step.
- **And it ends.** When both reserves reach zero, **Source** returns 0 forever, no guard on a positive quantum fires again, and the remaining tokens sit in metabolic channels. From there the only credit available is another organism’s stack. Proposition 44 is not an asymptotic remark about this listing; it is the last thing that happens in it.

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